

Explaining Network Intrusion Detection System with SHAP and LIME

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Abstract—Cybersecurity is a crucial field where data instances are continuously evolving as attackers attempt to exploit systems with novel attack patterns. Intrusion detection systems represent a key area for ensuring cyber safety. Machine learning and deep learning-based network intrusion detection systems have demonstrated promising outcomes. However, achieving higher accuracy often comes at the cost of increased complexity, which reduces interpretability. This lack of transparency makes it challenging to deploy such models in real-world settings, as the reasoning behind their decisions remains unclear. We apply data engineering techniques to enable the use of non-text data for LLMs. A multi-attention mechanism is applied to this data to convert it into a vectorized form, which is more effective for training because the embedded data is generated by capturing relationships, dependencies, and connections of each instance to all other data. We then fine-tune bidirectional RoBERTa models using this embedded dataset. More broadly, this approach demonstrates how datasets designed for traditional ML models can be reused for LLMs, reducing the need to create new datasets for LLMs. Finally, we utilize model explainability methods such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) to determine which features are more or less significant, thereby clarifying the rationale behind a decision. This also considers aspects such as which attributes influence the detection of a cyberattack and to what extent.

Index Terms—Explainable AI, LLMs, positional encoding, Multi-head attention;

I. INTRODUCTION

Distributed denial-of-service (DDoS) attacks represent a persistent danger in the realm of cybersecurity, dating back to the initial assault on the University of Minnesota in 1999 [1]. This incident involved a bandwidth exhaustion attack employing UDP flooding, which continued uninterrupted for two days. Such incidents were first documented in 1998 [2]. In the year 2000, numerous high-profile websites—including Amazon, eBay, and Yahoo—fell victim to Distributed Denial-of-Service (DDoS) attacks. Despite employing robust security measures to serve their customers, these platforms were compromised via the internet. These occurrences underscore the severity of DDoS as a critical menace that must be identified and mitigated to safeguard user access to data and services. The years 2015 and 2016 marked a particularly severe period,

with recorded attack volumes reaching 500 Gbps and 800 Gbps, respectively [3].

The growing prominence of the Internet of Things (IoT) is undeniable, with applications in automotive, wearable, and domestic spheres. Interconnecting these elements—from public infrastructure and home appliances to medical devices and self-driving cars—requires robust networking [4][5][6][7]. Scholarly work in this area includes the use of Support Vector Machines (SVM) for DDoS attack detection [8], and an algorithm developed by Dao et al. specifically for SDN-based devices [9]. Research into enhancing IoT security through Software-Defined Networking (SDN) is extensive. Contributions range from a distributed architecture for securing IoT in SDN domains [10] to analyses of vulnerabilities in the OpenFlow channel [11]. Further strategies for DDoS mitigation in IoT contexts have also been proposed. Also, other works present adaptive hypermedia models using XML, object-oriented design, and probabilistic user modeling, along with innovative OLAP techniques—such as data cube compression and handheld OLAP systems—to enhance personalization, mobility, and efficient data analysis [38-42].

Earlier researcher focuses statistical and machine learning models [13-16], quantum machine learning [17-18] with traditional dataset. While training model, data in an instance do not have any connection to other data based on dependencies, relations, semantic meaning, etc. This data make ML models difficulty to understand one data to other data. Traditional models that rely only on numerical or categorical features often fail to capture relationships, dependencies, or semantic meaning between data points within an observation. As a result, these models have limitations in learning deeper connections.

To address this, transformers are used on certain datasets. Transformers employ multi-head attention, where each attention head focuses on a different perspective of the input. For example, one head may capture dependencies, another semantic meaning, and another contextual relevance. Together, the multiple heads analyze tokens within an instance by examining their relationships and meanings. Ultimately, transformers generate a vectorized representation of the data that encodes its content and relationships. This provides a more

comprehensive picture, allowing one piece of data to connect meaningfully to another.

The seminal work "Attention is All You Need" presents the Transformer architecture, a model that fundamentally altered Natural Language Processing through its self-attention mechanism. This approach removes the need for recurrent structures and facilitates parallel computation, yielding superior results in applications such as text generation, translation, and related tasks [19]. Subsequent innovations, including BERT (2018) and GPT-3 (2020), significantly broadened these capabilities, ultimately leading to the emergence of 'foundation models' that have reshaped the landscape of artificial intelligence research [20-21].

Contemporary cybersecurity investigation has progressively centered on leveraging Large Language Models (LLMs) to enhance problem resolution [22–23]. Transformers employ numerous attention mechanisms distributed across 12 tiers to examine input sequences, whereas some studies fine-tune pre-trained LLMs like XLM-RoBERTa for specialized datasets [24]. Rahman et al. (2024) achieved 96.55% accuracy by combining multilingual BERT-produced embeddings with logistic regression to identify prompt injection assaults [25]. Expanding on this, we utilize multilingual BERT to produce code pattern embeddings to evaluate their efficacy in pinpointing vulnerabilities [26]. However, data deficiency continues to be an enduring obstacle when employing traditional embeddings [27]. This inquiry explores how contextual embeddings created by BERT can improve machine learning training in data-constrained environments for categorization tasks [28].

ML and DL algorithms have demonstrated remarkable effectiveness across numerous fields. Nevertheless, such achievements come with challenges in interpreting how these models operate, why a particular decision is made, which features or regions most strongly affect the output, and how confident the model is in its generated prediction. These and other concerns are often raised by end-users, especially when advanced techniques such as deep neural networks are employed. Consequently, a new research domain has emerged known as eXplainable Artificial Intelligence (XAI) which focuses on translating "black-box" models into more understandable forms [29]. XAI plays a crucial role in enhancing model interpretability and strengthening user confidence in its results [30-31]. Such assurances are vital for broader adoption of these models, particularly in sensitive and high-stakes domains like healthcare.

This manuscript is organized as follows: Section II presents Explainable AI (XAI), focusing on transparency and interpretability. Section III covers Large Language Models (LLMs). Section IV describes the dataset, and Section V details the methodology. Section VI delivers the results and discussion, leading to the conclusion which recaps the findings and spotlights future research directions.

II. EXPLAINABLE AI (SHAP & LIME)

Explainable AI (XAI) means developing artificial intelligence systems that can clearly show how and why they

make decisions instead of being "black boxes". This ensures that users can put their trust in the decisions made. By incorporating an interpretability layer into models, we can create more reliable and transparent systems that serve the interests of stakeholders. It helps to have a clear understanding of the model accuracy, transparency, fairness and outcomes. Explainable AI is crucial to build trust and confidence during deployment. Also, it helps to choose a responsible approach to AI development.

A. SHAP

SHapley Additive exPlanations (SHAP) is a comprehensive method in explainable AI (XAI) that allocates Shapley values to features, indicating their influence on an individual prediction of a machine learning model in comparison to the overall average outcome. It is model-independent, applicable to any type of machine learning algorithm, and offers localized explanations that can be combined to reveal broader, global patterns. SHAP aids in interpreting how features affect a model's results, allowing practitioners to build confidence in their systems and refine them by uncovering key feature significance as well as possible biases.

B. LIME

LIME (Local Interpretable Model-agnostic Explanations) is a prominent technique in the field of Explainable AI (XAI) designed to illuminate the reasoning behind individual predictions made by complex, opaque machine learning models—often referred to as "black boxes." Its core function is to explain a single prediction by approximating the complex model's behavior locally with a simpler, more transparent surrogate model, such as a linear classifier or a shallow decision tree. This approximation is focused specifically on the data point being examined. The LIME process unfolds through several key steps:

- **Perturbation:** LIME generates a new dataset by creating numerous slightly altered versions, or perturbations, of the original input instance. This is done by randomly modifying its feature values (e.g., setting certain features to zero or introducing minor variations).
- **Prediction & Weighting:** The original black-box model is then used to generate predictions for each of these new, synthetic data points. Each point is assigned a weight based on its proximity to the original instance; points closer to the original are considered more important. This proximity is typically calculated using a kernel function or a distance metric like cosine similarity for text or Euclidean distance for tabular data.
- **Surrogate Model Fitting:** A readily interpretable model is trained on this weighted, perturbed dataset. The goal of this simple model is not to replicate the global behavior of the complex model, but to faithfully mimic its decision boundaries in the immediate vicinity of the instance in question.
- **Explanation Extraction:** The parameters of this locally faithful surrogate model and the coefficients of models

become the explanation. These values quantify and rank the contribution of each input feature to the specific prediction, clearly identifying which factors were most influential for that particular case.

In essence, LIME answers the critical question: “For this specific input, which features were the primary drivers behind the model’s prediction?” By providing a localized, human-readable justification, it builds trust and accountability for AI systems.

III. LARGE LANGUAGE MODELS

Large Language Models (LLMs) is advanced deep learning based systems which is trained on extensive text datasets of text to process and generate human-like language. Leveraging deep learning structures, particularly the transformer model, they discern linguistic patterns to accomplish a variety of tasks. Multi-heads attentions mechanism of transformer enables parallel processing to find the most significant portions of the tokens to enhance its efficiency when handling large-scale datasets. We leverage encoder stacks to process tokens to calculate its attentions. A standard transformer model typically comprises an encoder and a decoder, which are responsible for compressing and reconstructing the input sequence data, respectively. The overall architecture is illustrated in Fig. 3.

Owing to the absence of a recurrent structure, which is essential for parallel computation, Transformers do not inherently manage sequential information. To address this, positional encoding is incorporated into each data taken as supplementary information which allows the model to recognize the absolute location of data within original sequence. The conventional sine and cosine functions are utilized for this encoding:

$$\begin{cases} PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000}\right), \\ PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000}\right), \end{cases} \quad (1)$$

These positional encodings are subsequently summed with the sequence embeddings to furnish distinctive identifiers for each location. This process enables the model to assimilate positional data. The self-attention mechanism, a cornerstone of the transformer design, augments sequence modeling by capturing interdependencies between elements, irrespective of their separation within the sequence. Its operation is founded on three parameter matrices: Query, Key, and Value. The query is the element currently being processed, while the keys function as identifiers to locate elements pertinent to that query. The values ($\mathbf{V}$) contain the actual information to be propagated upon a match. Score is computed by self-attention through assessing the similarity between $\mathbf{Q}$ and $\mathbf{K}$. A weighted sum is calculated by using this score. Multi-head self-attention mechanism execute multiple attention operations concurrently, permitting the model to glean insights from diverse representation subspaces. For every head i , attention is computed autonomously, and the outcomes are concatenated and linearly projected:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O, \quad (2)$$

where $\text{head}_i = \text{Attention}(\mathbf{QW}_i^Q, \mathbf{KW}_i^K, \mathbf{VW}_i^V)$.

This specialized architecture, by permitting simultaneous focus on various facets of the sequence, markedly strengthens the Transformer’s representational capacity, thereby boosting its efficacy and precision when managing intricate sequential data.

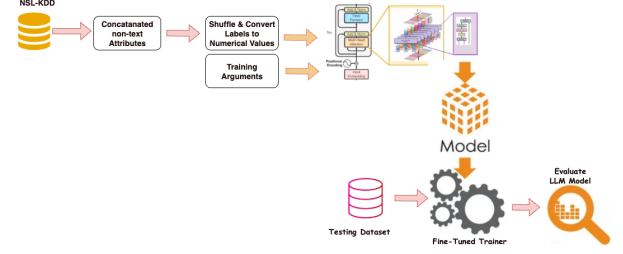


Fig. 1. An architecture of the proposed model for malware detection.

IV. DATASET

The NSL-KDD dataset was proposed to address several intrinsic limitations of the KDD’99 dataset. While this updated version still exhibits certain shortcomings [12] and might not perfectly reflect contemporary real-world networks, the absence of publicly available alternatives for network-based intrusion detection systems means it remains a valuable benchmark. It enables researchers to effectively evaluate and contrast diverse intrusion detection techniques. Additionally, the manageable volume of records in the NSL-KDD training and testing subsets allows for comprehensive experimentation using the entire dataset, eliminating the necessity for random sampling. This contributes to more uniform and reproducible evaluation outcomes across various studies.

NSL-KDD dataset provides the several merits compared to the original KDD dataset. It removes duplicated records from the training set which helps models not to be biased toward excessively common examples. The test sets contain no duplicate entries; therefore, the evaluation of learning models is not inflated by methods that perform well on repeated instances. The selected records from each complexity class is proportional inversely to their frequency in the original KDD dataset. As a result, the performance of various machine learning methods spans a wider variety, enabling more accurate and meaningful comparison of different algorithms. The size of datasets is practical which permits implementations without random sampling. This ensures that evaluation results are consistent and reproducible across studies.

V. METHOD

While training model, data in an instance do not have any connection to other data based on dependencies, relations, semantic meaning, etc. This data make ML models difficulty to understand one data to other data. Traditional models that rely only on numerical or categorical features often fail to capture relationships, dependencies, or semantic meaning between data points within an observation. As a result, these models

have limitations in learning deeper connections. To address this, transformers are used on certain datasets. Transformers employ multi-head attention, where each attention head focuses on a different perspective of the input. For example, one head may capture dependencies, another semantic meaning, and another contextual relevance. Together, the multiple heads analyze tokens within an instance by examining their relationships and meanings. Ultimately, transformers generate a vectorized representation of the data that encodes its content and relationships. This provides a more comprehensive picture, allowing one piece of data to connect meaningfully to another.

The proposed method uses a fine-tuned BERT model to detect intrusions in the NSL-KDD dataset by converting traditional network traffic features into a format suitable for large language model (LLM) processing. The workflow starts by taking all non-text attributes from each NSL-KDD record and merging them into a single text-like string. This simple transformation allows BERT—originally built for natural language—to interpret structured network data as a sequential input.

After the attributes are combined, the dataset is cleaned and prepared. Samples are shuffled to prevent any order-based bias, and categorical labels are converted into numerical values appropriate for classification. Key training parameters, such as learning rate, batch size, number of epochs, and optimization strategy, are selected to ensure stable and efficient learning. Once preprocessing is complete, the formatted data is fed into the BERT model for fine-tuning. During this stage, BERT’s transformer layers adjust themselves according to the patterns and behaviors found in cybersecurity data. Fine-tuning reshapes the model’s internal representations, enabling it to detect subtle anomalies, understand relationships among features, and distinguish normal traffic from malicious activity.

After training, the customized model is integrated into a dedicated training pipeline using an LLM-optimized trainer module. When the training phase finishes, the model is tested using the NSL-KDD test set to measure how well it classifies new, unseen traffic samples. Metrics such as accuracy, precision, recall, and F1-score are used to evaluate its overall performance. The final evaluation shows that the fine-tuned BERT model can reliably identify both attack patterns and normal traffic. Because it learns domain-specific behaviors during fine-tuning, it significantly outperforms traditional machine learning methods. Overall, this approach demonstrates that combining structured network features with advanced LLM techniques can greatly enhance intrusion detection performance.

VI. RESULTS AND DISCUSSION

We utilize Google Colab Pro’s computational resources to conduct our experiments, compensating for the hardware limitations of our local Mac with M2 chip (8GB RAM). This setup enables efficient processing for both model interpretability analysis and LLM fine-tuning tasks. Firstly, SHAP LIME provide model-agnostic interpretability, explaining why a model makes specific predictions by quantifying feature

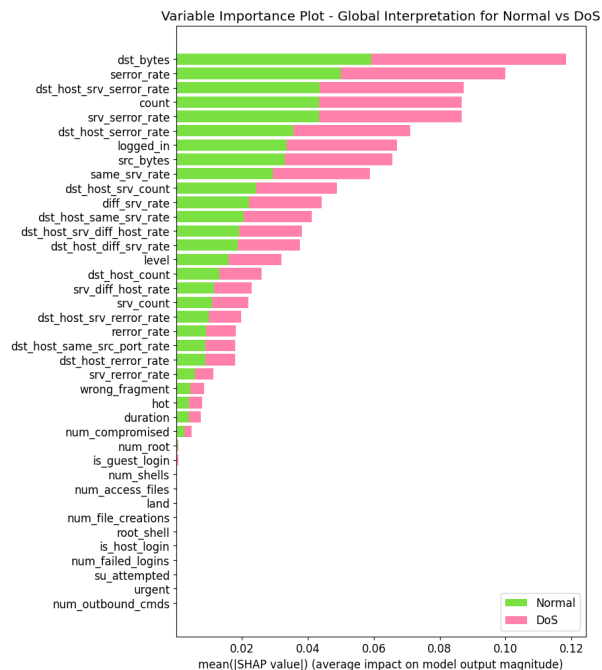


Fig. 2. SHAP-based feature importance highlighting influential attributes for distinguishing Normal versus DoS attacks.

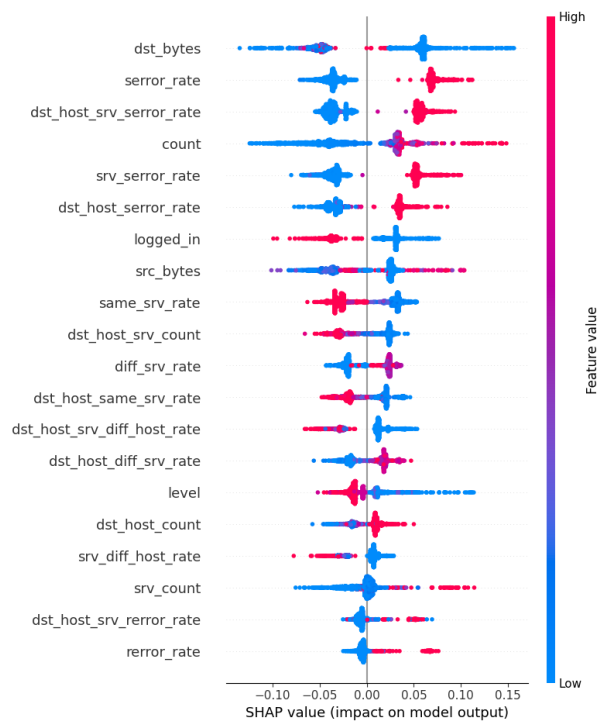


Fig. 3. SHAP summary plot showing feature contributions and impact magnitudes for Normal versus DoS classification.

importance for individual cases, moving beyond the "black box." Secondly, Fine-tuning LLMs for threat identification adapts a base model with specialized security data, enabling it to recognize nuanced patterns of malware, phishing, and vulnerabilities with high precision. Finally, we compare our works with existing works.

A. SHAP and LIME for Model Interpretability

The feature importance visualization offers a global perspective on the attributes that differentiate Normal from Denial-of-Service (DoS) traffic. The chart illustrates that both Normal (green) and DoS (pink) categories exert influence over numerous variables, with their contributions being relatively balanced across the two groups. Within this set, *dst_bytes*, *error_rate*, and *dst_host_srv_error_rate* distinctly stand out as the most significant factors, underscoring their strong role in discriminating between Normal and DoS instances (Fig. 2). These metrics are closely associated with anomalous network activity, including excessive data transmission and elevated error rates, which are hallmark indicators of DoS attacks. Conversely, attributes such as *num_outbound_cmds*, *is_host_login*, *land*, *urgent*, and *num_access_files* demonstrate negligible influence on the model's outcomes, implying their limited relevance for this specific classification objective. In summary, the evaluation underscores a concise group of pivotal features that primarily govern the predictive process.

The vertical axis displays the input attributes, ordered by their mean absolute SHAP value, positioning the most impactful predictors at the summit. The horizontal axis denotes the magnitude and direction of the SHAP values; positive values shift the model's output towards a DoS classification (label = 1), while negative values steer it towards a Normal traffic designation (label = 0). To illustrate, elevated values of *dst_bytes* (indicated by red data points) exhibit a strong correlation with Normal traffic. Conversely, diminished values (shown as blue points) influence the model to predict DoS. In a similar vein, high measurements for *error_rate* and *dst_host_srv_error_rate* elevate the probability of a DoS classification, and lower measurements diminish it. Variables such as *count* and *srv_error_rate* demonstrate a more complex, bidirectional influence, implying their effect on the final prediction is conditional and less straightforward (Fig. 3). To clarify these intricate relationships and investigate the interplay between features, dependence plots can be employed to yield a more profound understanding of the mechanisms behind the model's predictions.

Based on the provided LIME explanation for a specific instance classified as a Denial-of-Service (DoS) attack, the model's decision was primarily driven by a high volume of data sent to a destination (*dst_bytes* ≥ 451.00). This key factor positively contributed to the DoS prediction. Conversely, the absence of certain error-related features—specifically, both *srv_error_rate* and *error_rate* being equal to zero—exerted a strong negative influence on the prediction, meaning they significantly reduced the model's confidence in the DoS classification. Other attributes related to login attempts, fragmen-

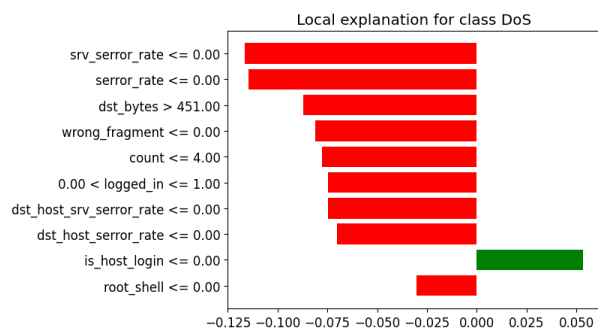


Fig. 4. A decision tree's local explanation for classifying a Denial-of-Service (DoS) cyber attack.

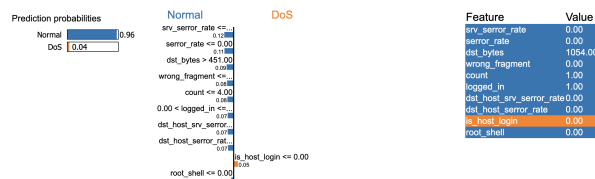


Fig. 5. SHAP visualization showing feature contributions for predicting Normal vs DoS network traffic.

tation, and connection counts had minimal to no impact on the final outcome for this specific instance (Fig. 4 and Fig. 5).

Based on the provided table, the description can be modified as follows: The output presents three key pieces of information: (1) the model's prediction probabilities, (2) the contribution of each feature to the prediction, and (3) the actual input value for each feature. We can observe that the network traffic instance is predicted to be Normal with 96% confidence. The primary reasons leading the model to this decision are:

- The *srv_error_rate* and *error_rate* are both 0.00, which strongly supported the Normal prediction.
- A high *dst_bytes* value (1054.00) was the third most influential factor, also contributing to the Normal clas-

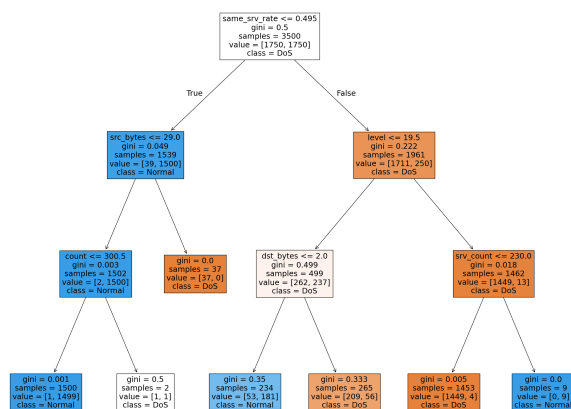


Fig. 6. Decision tree visualization showing feature-based splits for classifying Normal versus DoS network traffic.

sification.

These influential values, along with others such as `logged_in = 1.00`, can be verified in the table on the right.

By examining the tree structure, one can trace the decision-making process for each connection, providing insight into how the model distinguishes between Normal and Denial-of-Service (DoS) traffic. In the visualized tree, each node represents a split based on a specific feature's value. Internal nodes display the splitting feature and threshold (e.g., `same_srv_rate ≤ 0.495`), along with the Gini impurity as well as number of samples at that node. Leaf nodes show the predicted class (e.g., "class = DoS") and the distribution of samples. The color intensity in each node reflects the purity of the class—darker orange indicates a stronger association with DoS, while darker blue represents a stronger association with Normal. For example, a node with value = [1449, 4] and class DoS is intensely orange, indicating high confidence in predicting a DoS attack. Conversely, a node with value = [0, 9] is pure blue, confidently predicting Normal traffic. This visualization highlights the key features—such as `src_bytes`, `count`, and `dst_bytes`—that drive the model's classification decisions (Fig. 5).

B. Fine-tuning LLMs for identifying threats

The RoBERTa model was assessed on a dataset of 7,038 samples spanning 5 classes. Over three training cycles, the model demonstrated steadily strong results across all standard measures (Fig. 7). Accuracy enhanced from 0.9439 in the first epoch to 0.9779 by the third, while precision climbed from 0.9380 to 0.9783, indicating highly dependable positive classifications. Recall followed a comparable pattern, growing from 0.9439 to 0.9779, reflecting the model's capacity to accurately detect relevant instances. The F1-score, which harmonizes precision and recall, advanced from 0.9405 to 0.9754, confirming resilient and consistent classification effectiveness. In summary, RoBERTa successfully learned class distinctions in this multi-class task.

We evaluated the performance of the RoBERTa model on binary classification tasks involving Normal and DoS traffic using two datasets of different sizes. In the first experiment with 8,000 observations, the model achieved strong results across ten epochs. Accuracy, precision, recall, and F1 scores ranged between 0.9950 and 0.9975. After the third epoch, the performance stabilized, indicating that RoBERTa quickly converged and maintained reliability without significant fluctuations. This shows that the model is capable of effectively learning from relatively smaller datasets while maintaining consistency in its predictions (Fig. 8). In the subsequent experiment, using 20,000 observations, RoBERTa demonstrated improved performance. Across three epochs, the accuracy and other metrics consistently improved, reaching as high as 0.9987 by the third epoch. The uniformity across all four metrics—accuracy, precision, recall, and F1—demonstrates that the model not only generalizes well but also scales efficiently with larger training samples. The increase in dataset size enhanced the model's learning capacity, allowing it to

identify more nuanced patterns between normal and malicious traffic. Overall, these results highlight the robustness and adaptability of RoBERTa in intrusion detection tasks, confirming its suitability for cybersecurity applications where both accuracy and generalization are critically important.

The TABLE I presents a comparison of various existing models for intrusion detection, including Random Forest, LSTM, CNN-LSTM, ANN, and DNN. Existing approaches achieve good performance, with accuracy ranging from 86% to 97.93% and F1-scores between 86% and 97% [32-37]. While models like DNN show strong overall performance, certain metrics such as precision or recall are lower or unavailable for some methods, limiting their reliability. In contrast, the proposed RoBERTa-based approach consistently outperforms all existing methods, achieving 99.75% across accuracy, precision, recall, and F1-score. This demonstrates superior capability in capturing complex patterns and ensuring robust multi-class classification in intrusion detection.

However, the scalability of the proposed method is a valid concern in a big data context. Key improvements should include a detailed analysis of the computational cost and time of the data-to-text transformation and BERT fine-tuning stages. It is highly important to investigate the performance of this method on larger, real-world streaming datasets beyond NSL-KDD. Additionally, we will need more efficient transformer architectures (such as DistilBERT or DeBERTa) in future which can be explored and benchmarked against the current model to optimize the trade-off between accuracy and resource consumption.

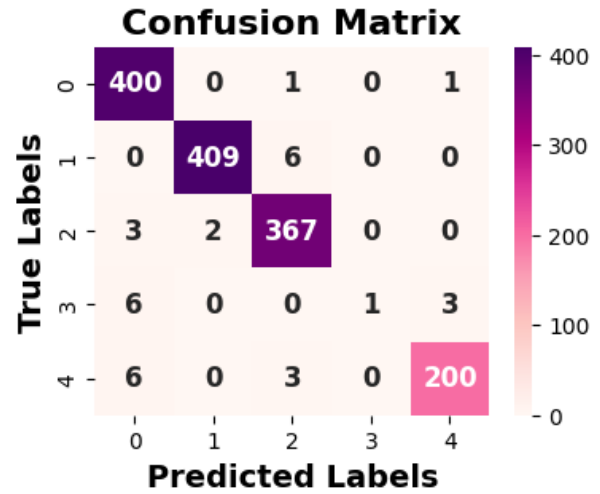


Fig. 7. Confusion matrix illustrating model performance in classifying Normal:0, DoS:1, Probe:2, U2R:3, and R2L:4.

VII. CONCLUSION

This study investigates the application of fine-tuned BERT and RoBERTa models for malware detection using the NSL-KDD dataset. Both models achieved a high accuracy of 99.75% when trained on 8,000 observations, demonstrating their effectiveness in identifying cybersecurity threats. Despite

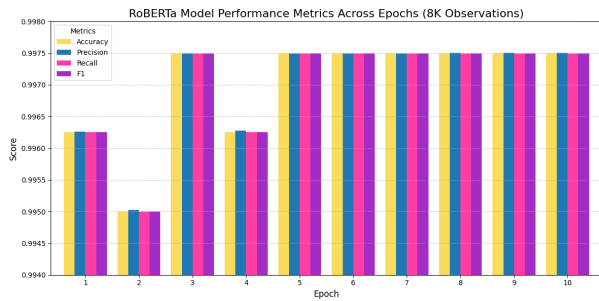


Fig. 8. Performance Progress by Fine-tuning RoBERTa based on Epochs

TABLE I
COMPARISON OF MODEL PERFORMANCE FOR INTRUSION DETECTION

Reference	Model & Features	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
32	Random Forest	86	89.5	86	86
33	LSTM	92.2	96	80	87
34	CNN-LSTM	87	85	88	86
35	ANN	93.75	78.56	92	87.42
36	ANN	94	N/A	N/A	N/A
37	DNN	97.93	97	97	97
Proposed	RoBERTa	99.75	99.75	99.75	99.75

their performance, the inherent complexity of such models necessitates interpretability analysis to ensure trust and transparency. Techniques including SHAP and LIME were employed to elucidate feature importance and model behavior across malware categories, providing global and local explanations for predictions. The results highlight the potential of combining advanced natural language processing models with explainable AI frameworks to develop reliable, interpretable, and trustworthy cybersecurity solutions.

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